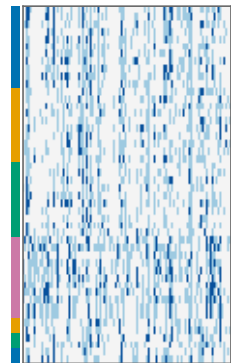
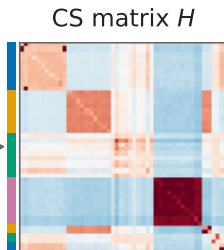


**detect-white: detect related pairs on the CS matrix, then whitening (default for large  $N$ )**

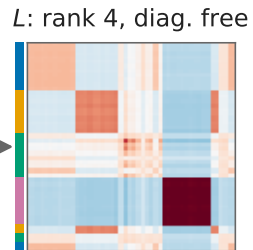
genotypes  $G$



one pass

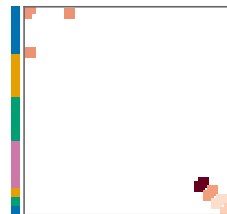


fixed rank  
+ kinship



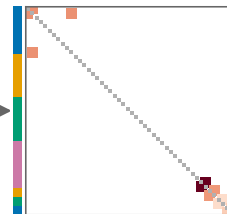
+

$S$ : pairs,  $\hat{\phi} > \tau$



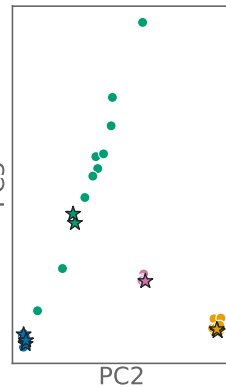
families,  
noise  $v$

$\Sigma$  (families)



whiten,  
 $\times \Sigma^{1/2}$

PCs (whitened)



● CEU ● CHB ● MXL ● YRI ★ relative